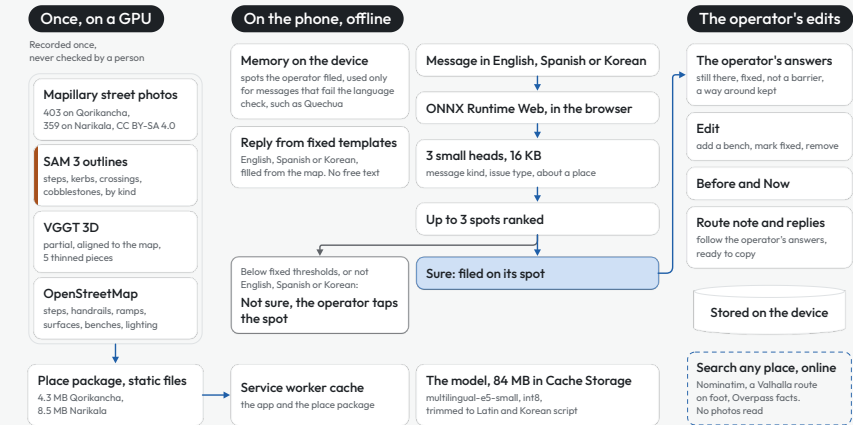


● Problem

Peru had 4,157,469 international visitors in 2025 (MINCETUR), and 71.7% of its employed people work in units of 1 to 10 people, 88.6% of them informally (INEI). A small tour operator gets visitor messages in languages the operator cannot read, about spots on a route that are hard to check. Quechua, the mother tongue of 3,735,682 Peruvians aged 5 and over (INEI, 2017 census), is not on our model's language list.

● Solution

An on-screen guide goes through the route one spot at a time. It shows the street photo where a model outlined steps or a kerb, what OpenStreetMap records there, and who it affects. The operator answers what is there now, can keep a way around that OpenStreetMap's router suggests, adds what the photos missed, and compares Before and Now. A small model on the phone reads each visitor message and names the spot, or says Not sure. Replies in English, Spanish or Korean come from fixed templates filled from the operator's answers. Every change needs the operator's tap.



Prepared once on a GPU; offline on the phone; the operator's edits on the device, and search, the one online part.

● Data

Qorikancha: 594 m in 60 stretches, 403 photos (2015 to 2023), 52 findings, 8 possible barriers at 5 spots, 44 OpenStreetMap records. A way around the steps is 679 m instead of 594.

Narikala: 1,020 m in 102 stretches, 359 photos (2016 to 2025), 49 possible barriers at 14 spots, 95 OpenStreetMap records.

None of these marks or records has been checked by a person.

● Prepared once, on a GPU

Photos: 403 public street photos for Qorikancha in Cusco and 359 for Narikala in Tbilisi, from Mapillary (a public street-photo site; CC BY-SA 4.0).

Marks: SAM 3 (an image model that outlines objects) marked steps, kerbs, crossings and cobblestones in the photos.

3D: VGGT (a model that builds 3D points from photos) made partial point clouds, aligned to the map and thinned to 5 pieces per route.

Map facts: OpenStreetMap (the open map) adds steps, handrails, ramps, surfaces, benches and lighting.

Package: each place ships as static files, 4.3 MB and 8.5 MB.

● On the phone, offline

Model: multilingual-e5-small (an open text encoder that turns a message into numbers), stored as 8-bit integers and trimmed to Latin and Korean script: 83,783,194 bytes, about 52 MB over the network.

Runtime: ONNX Runtime Web runs it in the browser. Three small trained classifiers (16 KB) give the message kind, the issue type and whether it is about a place, and the route's spots are ranked for each message.

Not sure: below fixed thresholds, or when a message does not look like English, Spanish or Korean, it says Not sure.

Offline: a service worker (the browser's offline store) keeps the app and the places. One message takes 25 to 38 ms in Chromium on the development Mac.

Search: search is the one online feature. It asks Nominatim, Valhalla and Overpass for a route on foot and OpenStreetMap's facts; no photos are read for it.

● Key results (synthetic test messages, held out by family)

Right spot first, held-out English, Spanish and Korean farm messages	46 of 48
The same on the Qorikancha route, with no new training	28 of 31
Exact keyword aliases: right spots on the route	3 of 34
Keyword aliases flag a place in praise, negations, resolved messages	15 of 15
Quechua (machine-translated), right spot first; 9 confident answers, 1 right	5 of 16
Language check: held-out Quechua answered Not sure, others unaffected	22 of 22
Quechua with 1, 2, 3 of the operator's filed messages per spot (preregistered, mean of 20 draws)	7.3, 8.8, 9.4 of 16

● Scalability

A new place needs public street photos and OpenStreetMap: it is prepared once and shipped as a package of a few MB. The model and its classifiers are shared across places. Search already gives any place a route from the map alone. Next: a review by native speakers, a field test with operators and a test on a low-cost phone.

● Responsible AI and limits

Messages, answers and edits stay on the device, and Start over deletes them. The model never writes text: it answers from fixed lists or says Not sure, and nothing changes without the operator's tap. Every test message is synthetic, written by a large language model, and none has been reviewed by a native speaker. No widths, heights, slopes or reachability are claimed, nothing has been measured on site, and no real operator has used it yet.